2026 HCI Prelim H1 Geography ANSWERS

1 A group of four Geography students in Singapore investigated the effectiveness of sheltered walkways as an adaptation strategy for residents coping with rising temperatures. They focused on a route between an HDB residential block next to their school and a nearby Mass Rapid Transit (MRT) station.

The students noted that while a sheltered walkway existed, it was incomplete as the walkway did not cover the entire route to the MRT station. Instead of walking to the MRT station from their homes, many residents continued to take a bus to the MRT station.

The students came up with the following geographical question for their fieldwork:

"Why do residents take a bus to the MRT train station instead of using the walkway?"

They divided themselves into two pairs, and conducted their fieldwork from 4 to 6 pm, every Wednesday in May as it was their designated home-based learning (HBL) day. Each day, one pair would count the number of residents using the sheltered walkway to get to the MRT station, while another pair would count the number of residents alighting from the bus at the MRT station. The students also noted the weather and traffic conditions.

In addition, they used a digital temperature sensor to measure the ambient temperature at seven selected points along the route. They also noted the features and details of the immediate surroundings at each point. They did this at 2 pm on a sunny Thursday afternoon in June.

Resource 1A is a map showing the walking route and the selected temperature measurement points. Resource 1B shows a photograph of a section along the route between Measuring Points 6 and 7. Resource 2 shows the number of residents using the route in the month of May, together with the prevailing weather and traffic conditions on each day. Resource 3A shows the information collected at each selected point along the route in June. Resource 3B shows a picture of the digital temperature sensor.

(a) Explain why the students' question is appropriate for their investigation. [3]

Award 1 mark for each explanation on why the students' question is appropriate for their investigation, to a maximum of 3 marks.

Point marked

- The research question is at a suitable scale and is manageable for a group of four students because:

  - the length of the route (measuring about 500 metres between the MRT station and the school) [1]
    OR

    the students have chosen to focus solely on a particular stakeholder group (residents)[1].

- The items in the research question are measurable, especially since the students are able to count the number of residents who prefer to walk and take the MRT [1].
- The research question is clearly defined, as seen from the:
  - clear identification of a stakeholder group (residents) that the investigation will focus on [1]
    OR
  - clear indication of the specific issue being investigated (i.e. that residents are continuing to take the bus instead of walking to the MRT station) [1].

(b) With reference to Resources 1A and 1B, explain how students can minimise potential risks when conducting their fieldwork along the walkway between their school and the MRT train station. [4]

Point marked
Award 1 mark for each explanation of how students can minimise potential risks when conducting their fieldwork along the route between their school and the MRT train station, to a maximum of 4 marks.

Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Possible responses include:
- Students should take care to give way to pedestrians, so as to ensure personal safety and minimise the occurrence of accidents [1]. For instance, students should take care not to bump into young children and elderly or obstruct those who are using Personal Mobility Aids (PMAs) [+1].
- Students should obey traffic signs and look out for traffic when crossing the roads at various junctions along the route [1]. This is to minimise the occurrences of accidents [1].
- Students should ensure that good walking shoes are worn, to minimise slipping/tripping [1]. This is especially so when walking on uneven and vegetated terrain beside the pavements/ avoiding tripping and falling into canal [+1].
- Students should ensure that they always work in pairs or in a group so that they can look out for one another's safety [1].
- Hydration/ sunscreen to mitigate heat stress [1]

(c) Suggest reasons for the number of residents walking to the MRT train station shown in Resource 2. [4]

Point marked. Award 1 mark for each explanation of how weather conditions may have influenced the number of students using the sheltered walkway, to a maximum of 2 marks.
Award 1 mark for each explanation of how traffic conditions may have influenced the number of students using the sheltered walkway, to a maximum of 2 marks.
Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Possible responses include:
- The occurrence of heavy rain on 3 May likely resulted in fewer residents using the walkway to get to the MRT station because parts of the walkway are not covered/sheltered [1]. This may have led residents to take the bus instead to stay as dry as possible [+1] or avoid the hassle of bringing/ forgetting to bring an umbrella [+1]
- Very heavy traffic congestion due to roadworks on 17 May could have resulted in more residents using the walkway to get to the MRT station because bus services and private cars may have been delayed [1]. As a result of this, it may have been faster for the residents to walk and take the MRT instead [+1].
- Cooling weather on a cloudy day – residents may not mind saving bus fare/ walking to get some exercise [1]

(d) With reference to Resources 1A, 1B, and 3A, explain why the students' selection of the 7 temperature measurement points is appropriate. [5]

Point marked. Award 1 mark for each explanation of why the students' selection of the 7 temperature measurement points is appropriate, to a maximum of 5 marks.
Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Possible responses include:
- The sample size (i.e. number of measurement points) is manageable for a group of 4 students to complete in one afternoon [1].
- Measurement points are sufficiently far apart from one another along the route, which improves the representativeness of the sample [1].

- Measurement points are differentiated by their coverage from the sheltered walkway, which is relevant to the students' interest in finding out the effectiveness of sheltered walkways [1]. For example, 5 of the measurement points are covered by the sheltered walkway, and 2 of them are not covered [+1].
- Measurement points are also differentiated by their surrounding features, which helps students recognise the possible influence of other factors on the walkability of the walkway [1]. For example, the surrounding features observed by the students included a canal, varying densities of planted trees, and varying types of vegetation [+1].

(e) With reference to Resources 3A and 3B, explain how the students can ensure the reliability of the ambient temperature data they collected. [4]

Point marked. Award 1 mark for each explanation of how the students can ensure the reliability of the ambient temperature data they had collected, to a maximum of 6 marks.
Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Possible responses include:
- Students can repeat the measurements three times and take an average reading at each measurement point [1].
- Students can hold the digital temperature sensor consistently at the same height at all measurement points, when taking the readings [1].
- Students should handle/use the digital temperature sensor correctly, to minimise inaccurate and inconsistent readings of the temperature [1]. For example, students should ensure that the sensitive stainless steel probe on the digital temperature sensor is not in contact with or blocked by the hand when taking the reading [+1].
- Students should be familiar with operating the digital temperature sensor, such as setting it to the correct modes, to ensure the accuracy and consistency of their measurements [1]. For example, they should ensure the sensor is measuring ambient and not infrared temperature at all times or ensure the temperature readings are consistently measured using the same unit of measurement (i.e. oC or oF) at all times [+1]

(f) The students concluded that residents do not use the sheltered walkway mainly because it did not fully cover the route between their homes and the train station. With reference to all the resources, evaluate the validity of the students' conclusion. [10]

Levels marked according to the H2 Generic Levels Descriptors for the Fieldwork Evaluative Question.

Candidates could refer to Resources 2 and 3B and consider how far the collected data can support the students' conclusion. Candidates could acknowledge how the data suggests that higher ambient temperatures under the unsheltered sections of the route may have reduced its walkability. Moreover, candidates could observe how the walkway was least used during heavy rainfall, suggesting how unsheltered parts of the walkway may have deterred students from using it during rainy weather.

Candidates could also look at the way data was collected and assess if the methods used could compromise the accuracy and reliability of the data. For instance, they could look at whether the collection of data over a few days at one timing and in only one part of the year/ during their HBL days is sufficient in providing robust data for their investigation.

Additionally, candidates could consider what other data may be needed for this investigation. For instance, they could mention the need to collect qualitative data and find out whether there are other factors (such as the perceived convenience of other modes of transport) that deter residents from walking to the MRT station.

- Students should note higher temperatures at Point 7 compared to Point 6. Reasons include the lack of shelter (Resource 1B) and the presence of heat-absorbing concrete surfaces or lack of vegetation (Resource 3A), leading to higher local ambient heat

- The investigation has low validity because the "Why" question is not answered by temperature and count data alone. The timing is also inconsistent (May for counts, June for temperatures).
- Low reliability as temperature data was only collected once (one Thursday), making it a "snapshot" that doesn't account for diurnal or day-to-day variability.
- Improvements: Split into pairs to take simultaneous measurements at sheltered and unsheltered points; increase sample size to multiple days; include qualitative interviews

2 Resource 4A shows different mobility patterns between women and men in Vienna. Resource 4B shows different activities of the unpaid reproductive work performed by women and men in Vienna. Resource 5 shows the level of satisfaction of women living in Vienna with their living situation in 2022. Resource 6 shows Vienna's measures for a 15-minute city.

(a) Describe the distribution of mobility patterns and activities of unpaid reproductive work between women and men shown in Resources 4A and 4B. [4]

For mobility patterns, women disproportionately travel more on foot and public transport than men. [1]. However, this pattern is reversed for travel by bike and car. [+1]

For unpaid reproductive work, women's involvement in such activities is significantly greater than men for every category. [1]

Women are involved at least 65% and up to 78% of the time while men are involved at the most 35% with taking children to kindergarten at the lowest of 22%. [1]

(b) With reference to Resource 5, cite data to compare the level of satisfaction of women living in Vienna with the aspects of their living situation. [4]

Similarity:
Women are generally satisfied with their living situation with satisfaction with public transport going up to 99% and the lowest is leisure facilities at 85%. [1]

Differences:
Satisfaction with public transport is doing the best with 62% of women completely satisfied, while leisure activities performs the worst with only 23% [1] The bulk of the responses for leisure activities fall under very satisfied and moderately satisfied at 28% and 25%. [+1]

Public transport, green areas and safety have completely satisfied as highest % as compared to neighbors, public spaces and leisure facilities which have lower completely satisfied % than moderately satisfied. [1]

(c) Explain how gender equality contributes to progress towards liveable cities. [4]

Possible content:

- Gender equality encourages planners to recognise that women may experience urban spaces differently from men. Appropriate urban design increases women's ability to access education, employment, recreation and social activities. [1] When there is a greater perception of and actual safety, there is greater freedom to use urban spaces leading to an improved quality of life. [+1] This can involve:
    - Better street lighting, CCTV, emergency facilities and safer public transport can reduce women's fear of crime. [+1]
    - Designing streets with good visibility and active public spaces can make women feel safer [+1]

- Women often have different mobility patterns because they may combine employment with childcare, shopping and caring for elderly family members.[1] Gender-sensitive transport planning can therefore provide frequent and reliable public transport; accessible routes for women travelling with children. [+1] This reduces the time and financial costs of mobility, allowing women greater access to employment and services leading to greater economic and social well-being. [+1]

- Gender equality can increase women's participation in the urban economy such as equal access to education and employment, protection against workplace discrimination and affordable childcare. [1] Greater economic participation can increase household incomes and women's economic independence improving social and economic well-being. [+1]

(d) Suggest how the concept of a 15-minute city and micromobility shown in Resource 6 may address the needs of women in Vienna shown in Resources 4A, 4B and 5. [6]

Possible content:
Locating shops, schools, parks and other services within a short distance reduces the need for long journeys allow for safe, attractive pedestrian areas can make walking more practical for everyday trips. [1] Urban greening and urban seating make streets more pleasant and provide places to rest. [+1]

Having services close to transport nodes reduces walking distances and makes journeys involving children or shopping easier. [1] Mobility hubs bring different forms of transport together, making transfers between modes easier, hence women can combine walking with public transport for longer journeys. [+1]

The bike lane, parking racks and mobility hubs support micromobility. Bikes, e-bikes and potentially scooters allow women to cover short distances quickly without relying on a private car. [1] This is particularly useful when making several short trips in one journey as in the case of trip chaining. [+1]

(e) Explain why the success in urban reimaging strategies may vary across places. [4]

Possible content:
Cities in more developed countries have greater ability to fund large-scale regeneration, infrastructure and public spaces. [1] Limited government finances may constrain the scale and quality of redevelopment in less developed cities. [+1]

Involving multiple stakeholders such as local businesses and residents through consultation and participatory planning can improve the social acceptability and inclusiveness of the project. [1] If decision-making is dominated by governments or developers, projects may prioritise economic growth, tourism and investment over residents' needs. [+1]

Success of reimaging strategies also depends partly on whether lower-income groups can remain in and benefit from the regenerated area as they are often displaced when changes take place. [1] Strategies that address the needs of the local residents tend to be more successful such as providing affordable housing, improved public transport, provision of public spaces and community facilities, training and employment opportunities for local residents and maintaining affordable shops and services. [+1]

3 Resource 7 shows Antarctic daily sea ice extent 1981-2010 average, 2023 and 2025. Resource 8 shows dominant drivers of tree loss by region, 2001 – 2025. Resource 9 shows the process of melting ice sheets.

(a) Cite data to describe the Antarctic daily sea ice extent for 2025, 2023 and 1981-2010 average shown in Resource 7. [4]

The 2023 and 1981 – 2010 values reflect very similar trends except that overall, the 2023 values are lower by about 2 million km2 [1] Both dip from Jan to March, with the lowest value for March at 2 million km2 for 2023 [+1]. Both increase from Mar to Oct, peaking in Oct at 18.5 million km2 and 16.5 million km2 in 1981 – 2010 and 2023 respectively. [+1]

The 2025 values are different as between Jan and Mar, it dips at a faster rate, falling by 5 million km2 compared to 4 million km2 for 1981- 2010 and 3 million km2 for 2023. [1]

(b) Suggest reasons for the Antarctic daily sea ice extent in 2023 shown in Resource 7. [4]

The overall pattern is the result of seasonal changes in temperature between winter and summer. [1] The lowest extent occurs in summer/spring between Nov – Mar as it is warmer and the sea melts. [+1]

The 2023 values are lower than that of 1981-2010 due to contemporary climate change as mean global temperatures are higher. [1] Greenhouse gas concentrations have increased, absorbed more heat, caused melting and the overall extent of Antarctic sea ice to be lower. [+1]

(c) Compare the dominant drivers of tree loss by region shown in Resource 8. [4]

Overall, all the regions have a mix of several drivers of tree loss, although some drivers dominate more than others. [1]
Wildfires dominate North America, Oceania and Asia with at least half of the tree loss caused by it. [1]
Permanent agriculture clearly dominates as the cause of tree loss in Latin America and SEA with almost 75%. [1]
While logging in North America and Asia has a relatively large influence of about 25 – 30%, it features most significantly in Europe with at least 80%. [1]

(d) With reference to Resources 8 and 9, explain how positive feedback may accelerate warming. [6]

Deforestation removes trees, reducing the amount of vegetation available to absorb CO_2 through photosynthesis. [1] Trees that are cut down may be burned or decomposed, releasing stored carbon as CO_2. Higher atmospheric CO_2 strengthens the greenhouse effect, contributing to warming. Warmer and drier conditions in some regions can increase drought, wildfire and forest dieback. Further loss of forest means even less CO_2 is removed from the atmosphere. [+1]

Large ice sheets have a high albedo, meaning they reflect a large proportion of incoming solar radiation. Global warming causes ice sheets to melt, exposing darker surfaces such as rock, soil or ocean water. [1] These darker surfaces have a lower albedo and absorb more incoming solar radiation. More solar energy is converted into heat, causing further warming. Further warming causes more ice to melt, exposing still more dark surfaces. [+1]

Melting of large ice sheets cause the glacier to thin and their surface sinks to lower altitudes where temperatures are higher. [1] As a result, there is greater melting of ice causing the glacier to thin further and sink to even lower altitudes, experience higher temperatures which lead to even more melting. [+1]

(e) Explain why climate change in the past 200 years is likely caused by human activities. [4]

There is overwhelmingly clear evidence that climate change in the past 200 years is caused by human activities as increases in greenhouse gas emissions and concentrations by industrial activity and vehicles are closely associated with the spike in mean global temperatures as reflected by the Keeling Curve and temperature graphs. [1] Increased GHG concentrations absorb more longwave radiation in the atmosphere and this reradiated back to the Earth's surface where it is being absorbed, resulting in higher temperatures. [+1]

The rate of warming is occurring much faster than most natural climate changes caused by factors such as volcanic eruptions, solar variations and changes in Earth's orbit. However, the rate of recent warming is unusually rapid. [1] The IPCC estimates that global surface temperature in 2010–2019 was about 1.07°C warmer than 1850–1900 due to human influence, while natural factors contributed only about −0.1°C to +0.1°C over the same period. [+1]

4 'Psychological well-being is the most challenging dimension to achieve when creating liveable cities for the elderly.'

Evaluate this statement. **[13]**

This question requires students to weigh the three dimensions of well-being for the elderly listed in the syllabus: economic, social, and psychological well-being. It also connects to the subjective nature of liveability, which is influenced by time, place, and the assessor. They should also consider how the effectiveness of strategies to create liveable cities for the elderly vary across countries at different levels of development.

Possible approach
Students should compare the relative difficulty of addressing tangible issues, such as economic well-being (poverty or lack of retirement funds), with intangible issues like psychological well-being (loneliness or loss of purpose). A strong response will evaluate how the primary "challenge" may shift depending on the urban context, and likely vary across space the time.

5 'It is impossible for effective climate change response to be unilateral in nature.'

Evaluate this statement. **[13]**

Possible content:
Contemporary climate change is a collective-action problem because its causes and consequences cross national borders and involve almost every level of society. Greenhouse gases emitted in one country contribute to global atmospheric warming, while the effects—such as sea-level rise, extreme weather and ecosystem disruption—are experienced across borders. Consequently, no single government, business, NGO or community can adequately address climate change alone. Effective responses require coordination between the state, NGOs, corporations and communities, with each contributing different resources and forms of influence. The state can provide the essential authority and coordination, NGOs contribute advocacy and accountability, corporations provide investment, innovation and changes in production, and communities provide local implementation, adaptation and social participation.

Hence, the response to contemporary climate change cannot realistically be a unilateral effort. Its global scale means that action by one country cannot prevent emissions or impacts elsewhere, while its complexity means that government action alone cannot transform production, consumption and everyday behaviour. The most effective response is consequently not a choice between these actors but a coordinated system in which their different capacities reinforce one another and fill the gaps that other actors cannot fulfil. The challenge is to establish institutions and policies that make such cooperation possible while ensuring that responsibility for climate action is not shifted from powerful actors onto communities and individuals alone.

6 Evaluate the influence of high urbanization rates on contemporary climate change. **[13]**

Possible approach:
Candidates can approach the question by making a judgment of the extent to which high urbanization rates influence contemporary climate change. Candidates can consider how activities in urban areas such as industrial activity, transportation, production of goods and services and lifestyle of urban population may contribute to high intensity use of energy and the production of greenhouse gases. In turn, the increasing concentration of GHGs are the main cause of contemporary climate change. Candidates should also consider other possible reasons for contemporary climate change such as the destruction of carbon sinks through human activities such as deforestation and destruction of coral reefs. Natural causes of climate

change such as sunspot activity and the Milankovitch cycles can also be brought into the discussion to consider the extent and relevance of influence. Criteria to assess the extent of influence include the spatial and temporal scale of the influence of the factors.